(c) Nabinagar Power Plant - Indian Railways
(d) Kayamkulam Power Plant - National Thermal Power Corporation

I.A.S. (Pre) 2005

Ans. (a)

Integrated steel plant in Kalinganagar Industrial Complex in Jajpur district of Odisha, is working under Tata Steel not Steel Authority of India.

21. The Barh Super Thermal Power Station (BSTPS) is located in which State?

(a) Bihar (b) Karnataka
(c) Rajasthan (d) Punjab
(e) None of the above/More than one of the above.

63rd B.P.S.C. (Pre) 2017

Ans. (a)

Barh Super Thermal Power Station is located in the Patna district of Bihar state. It has a total approved capacity of 3300 MW and an installed capacity of 1320 MW. The Unit 4 was commissioned in the year 2014 and Unit 5 in the year 2016.

22. Which of the following is not correctly matched?

(Station)	(Power)
(a) Patratu	Thermal
(b) Jhakri	Hydroelectric
(c) Kalpakkam	Nuclear
(d) Korba	Wind

U.P.R.O./A.R.O. (Mains) 2016

Ans. (d)

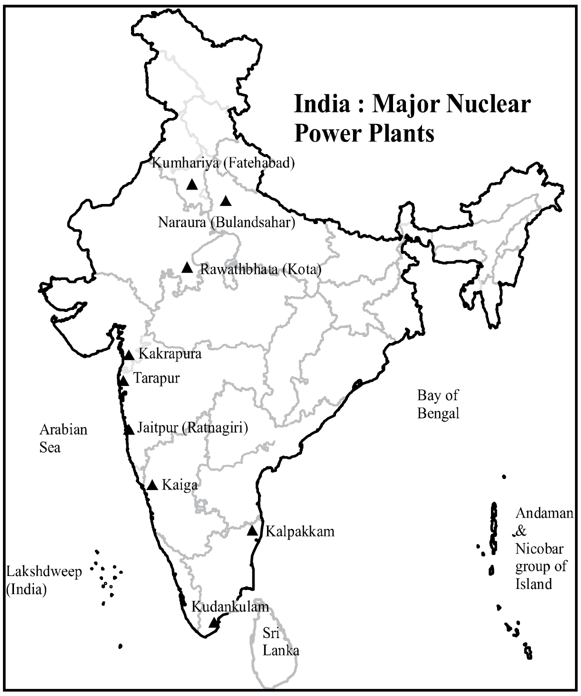
The matching of the centers in question and their respective power plants is as follows :

(Station)	**(Power)**
Patratu	Thermal Power Station
Jhakri	Hydroelectric Station
Kalpakkam	Nuclear Power Station
Korba	Thermal Power Station
Hence option (d) is not correctly matched	

ii. Nuclear Energy

*The most promising source of energy to fulfill the future demand for energy in India is nuclear energy. India's **first Nuclear Power Reactor** was established at Tarapur in 1969. Tarapur Atomic Power Station was constructed initially with a boiling water reactor (BWR) that was based on American technology. *Production of nuclear energy requires **uranium thorium** and **heavy water**. *Uranium is the most widely used mineral in Atomic Power Station as a fuel.

*A total of around 462001.66 MW of energy capacity has been installed in the country as of 31 December, 2024 which includes 8180 MW from nuclear energy which is 1.77% of the total installed energy capacity. ***Rawatbhata Atomic Power Station** was established in 1973 with the cooperation of Canada in Kota.



***Narora Atomic Power Station** is located in Narora, Bulandshahar District in Uttar Pradesh. ***Kakrapar Atomic Power Station** is located in Surat District of Gujarat. *In 1988, **Kundankulam Nuclear Power Plant** was established with the cooperation of Russia (the then Soviet Union) in Kundankulam, Tirunelveli District of Tamil Nadu through an agreement. Under this agreement, Russia has been building two nuclear reactors for India since 2002. Under the Intergovernmental Agreement in December 2008, Russia agreed to supply four more nuclear reactors to India. All four reactors are in the implementation phase. ***Kaiga Generating Station** is a nuclear power generating station situated at Kaiga, near the river Kali, in Karnataka. It has four units.

*The First Reactor of the Kundankulam nuclear power plant, which is located in Kundankulam, of Tamil Nadu, is a pressurized water reactor. ***Mithivirdi Nuclear Power Station** was established with the cooperation of the U.S.A. The project is a part of the Indo-U.S. nuclear deal signed in 2008. In 2015, the 6000 MW project was approved by the Ministry of Environment, Forest and Climate Change for the coastal regulatory area. This nuclear power plant is located in the Bhavnagar district of Gujarat. ***Jaitpur Nuclear Power Station** in India is proposed in cooperation with Areva Company of France in Ratnagiri, Maharashtra. For the nuclear energy, a heavy water plant is required. ***Thal Heavy Water Plant** is located in Maharashtra. ***Manuguru Heavy Water Plant** is located in Telangana. *There are seven heavy water plants in the country. These are Vadodara (Gujarat), Hazira (Gujarat), Kota (Rajasthan), Thal (Maharashtra), Manuguru (Telangana), Talcher (Odisha) and Tuticorin (Tamil Nadu). *Nuclear Power Corporation Limited is the joint venture of NTPC and NPCIL. *NPCIL has a **51%** stake in Nuclear Power Corporation Limited and **49%** of NTPC. It was established in January 2011.

1. In which place was the first Nuclear Power Station established in India?

(a) Kalpakkam (b) Kota
(c) Tarapur (d) Narora

Chhattisgarh P.C.S. (Pre) 2014

Ans. (c)

Tarapur Atomic Power Station (TAPS) is located near Boisar in Thane district of Maharashtra. It is the first Nuclear Power Station of India which commenced its commercial operation in 1969.

2. In the year 2006-2007, the share of nuclear energy generated in India in the total energy generated was –

(a) Less than 3 percent
(b) Between 3 and 4 percent
(c) Between 4 and 6 percent
(d) Between 6 and 8 percent

U.P.P.C.S. (Mains) 2007

Ans. (b)

When the question was asked option (b) was correct answer. According to National Power Portal data as on 31 December, 2024, nuclear power capacity was 8180 MW i.e. 1.77% of the total installed energy capacity (462001.66 MW). Thus option (b) would be correct.

3. The share of nuclear energy generated in India in the total energy generated approximately is :

(a) 2% (b) 3%
(c) 4% (d) 5%

U.P.U.D.A./L.D.A. (Pre) 2013

Ans. (b)

See the explanation of the above question.

4. India is self-sufficient in the supply of –

(a) Uranium (b) Thorium
(c) Iridium (d) Plutonium

U.P.P.C.S. (Mains) 2005
U.P. Lower Sub. (Pre) 2002
U.P.P.C.S. (Pre) 2003

Ans. (b)

Uranium is the most significant nuclear mineral used as nuclear fuel. Thorium is extracted mostly from monazite.

5. Match List-I with List-II and select the correct answer using the codes given below the lists:

List- I Nuclear Power Station	List- II States
A. Kota	**1. U.P.**
B. Tarapur	**2. Gujarat**
C. Kakrapara	**3. Maharashtra**
D. Narora	**4. Rajasthan**

Code :

	A	B	C	D
(a)	1	2	3	4
(b)	4	3	2	1
(c)	3	4	1	2
(d)	3	2	1	4

U.P.P.C.S. (Pre) 1999

Ans. (b)

According to the given options, the correct matching of nuclear power stations and concerned states is as follows :
Rawathbhata (Near Kota) Atomic Power Station - Rajasthan
Tarapur Atomic Power Station - Maharashtra
Narora Nuclear Power Station - Uttar Pradesh
Kakrapara Nuclear Power Station – Gujarat
Thus, option (b) is correct.

6. Match List-I with List-II and select the correct answer using the codes given below the lists:

List- I (State)	**List- II (Atomic Power Station)**
A. Gujarat	**1. Narora**
B. Karnataka	**2. Kakrapara**
C. Rajasthan	**3. Rawatbhata**
D. Uttar Pradesh	**4. Kaiga**

Code :

	A	B	C	D
(a)	1	2	3	4
(b)	4	3	2	1
(c)	2	4	3	1
(d)	4	2	1	3

U.P.U.D.A./L.D.A. (Pre) 2002

Ans. (c)

See the explanation of the above question.

7. Which one of the following pairs is correctly matched?

Nuclear Plant	**Year of Commissioning**
(a) Kota	1989
(b) Kakrapara	1984
(c) Kaiga	1999
(d) Kalpakkam	1995

U.P.P.C.S. (Mains) 2007

Ans. (*)

The year of commissioning of the nuclear plants given in the question is not correctly matched. The year of commissioning of nuclear plants is Kota -1973, Kakrapara-1993, Kaiga-2000 and Kalpakkam - 1984.

8. Consider the following statements:

Assertion (A): Nuclear energy is a promising source of futuristic demand for energy supply in India.

Reason (R) : Nuclear minerals are ubiquitously available in India.

Select the correct answer by using the codes given below:

(a) Both (A) and (R) are true, and (R) is the correct explanation of (A).

(b) Both (A) and (R) are true, but (R) is not the correct explanation of (A).

(c) (A) is true, but (R) is false.

(d) (A) is false, but (R) is true.

U.P.U.D.A./L.D.A. (Pre) 2006

Ans. (c)

Nuclear energy is a promising source of futuristic demand for energy supply in India. Although uranium is available in small quantities in India, it is dependent on imports for it. Although thorium is found in abundance in India, the molecule is not available everywhere in India. According to National Power Portal data as of 31 December, 2024 total nuclear power capacity was 8180 MW i.e. 1.77% of the total installed power capacity.

9. Match List-I with List-II and select the correct answer using the codes given below the lists:

List- I (Atomic Power Plants/ Heavy Water Plant)	**List- II (State)**
A. Thal	**1. Andhra Pradesh**
B. Manuguru	**2. Gujarat**
C. Kakrapara	**3. Maharashtra**
D. Kaiga	**4. Rajasthan**
	5. Karnataka

Code :

	A	B	C	D
(a)	2	1	4	5
(b)	3	5	2	1
(c)	2	5	4	1
(d)	3	1	2	5

I.A.S. (Pre) 2005

Ans. (d)

Thal	–	Maharashtra
Manuguru	–	Andhra Pradesh (now in Telangana)
Kakrapara	–	Gujarat
Kaiga	–	Karnataka

10. Which one of the following is not correctly matched?

(a) Kaiga - Karnataka

(b) Rawat Bhata - Rajasthan

(c) Muppandal - Tamil Nadu

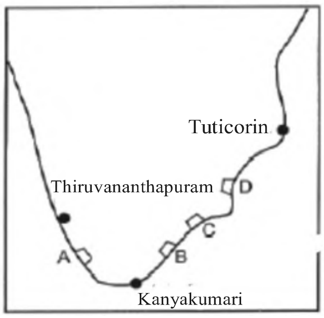
(d) Ennore - Meghalaya

U.P.P.C.S. (Mains) 2015

Ans. (d)

In the given option, Ennore is situated in Tamil Nadu, not in Meghalaya. The other options are correctly matched.

11. **The given figure shows a portion of Southern India. The proposed site Kudankulam for the construction of two 1000 MW nuclear power plants has been labeled in the map as –**



(a) A (b) B
(c) C (d) D

I.A.S. (Pre) 1999

Ans. (b)

Kudankulam Nuclear Power Plant is a nuclear power station in Kattabomman in the Tirunelveli district of Tamil Nadu which is located on the map by point B.

12. **Kudankulam Nuclear Power Plant is being established in –**

(a) Tamil Nadu
(b) Kerala
(c) Andhra Pradesh
(d) Karnataka

U.P.P.C.S. (Mains) 2011

Ans. (a)

See the explanation of the above question.

13. **Russia has agreed to install how many units of nuclear reactors in Kudankulam Nuclear Power Plant?**

(a) 02 (b) 04
(c) 05 (d) 06

U.P. Lower Sub. (Pre) 2004

Ans. (d)

An Inter-Government Agreement (IGA) on the project was signed in 1988 by the then Soviet head of State Mikhail Gorbachev, for the construction of two reactors to be installed at Kudankulam Nuclear Power Plant in Tamil Nadu.

14. **India is constructing its 25th Nuclear Plant at –**

(a) Bargi (Madhya Pradesh)
(b) Fatehabad (Haryana)
(c) Kakrapar (Rajasthan)
(d) Rawat Bhata (Rajasthan)

U.P.P.C.S. (Mains) 2011

Ans. (d)

The 25th Nuclear Power Plant of India is located at Rawatbhata in the State of Rajasthan where its construction began in July, 2011.

15. **The twentieth nuclear power station of India is**

(a) Tarapur
(b) Rawatbhata
(c) Kaiga (Karnataka)
(d) Narora (U.P.)

U.P. Lower Sub. (Pre) 2013
U.P.U.D.A./L.D.A. (Spl) (Pre) 2010

Ans. (c)

The twentieth nuclear power station of India is located in Kaiga, Karnataka.

16. **Which of the following places does not have heavy-water plant for atomic energy?**

(a) Kalpakkam
(b) Hazira
(c) Thal
(d) Tuticorin
(e) Manuguru

Chhattisgarh P.C.S. (Pre) 2013

Ans. (a)

Heavy water plants for atomic energy are located in Hazira (Gujarat), Vadodara (Gujarat), Kota (Rajasthan), Manuguru (Telangana), Talcher (Odisha), Thal (Maharashtra) and Tuticorin (Tamil Nadu). Hence (a) is the correct answer.

17. **'Mithivirdi' Nuclear Power Plant will be set up in collaboration with which of the following countries?**

(a) USA (b) Canada
(c) Russia (d) France

U.P.R.O./A.R.O. (Pre) 2014

Ans. (a)

Mithivirdi Nuclear Plant will be set up in collaboration with the United States of America (U.S.A.). It was proposed to be set up in Talaja Taluka, Bhavnagar, Gujarat. The project is part of the Indo-U.S. nuclear deal signed in 2008.